

① Lab 1 → Python Fundamentals.

② Lab 2 → Intro to Data Analysis. → ^{scikit-learn Dataframe} Linear Regression
Logistic Regression
Decision Trees.

Lab 3 → ~~Decision Trees~~ Cuparis + Algorithm Bias. I → Decision Trees.

Lab 4 → ~~Perceptron~~ + Multi-Algorithm Comparison II → Random Forest
SVM
KNN

Lab 5 → Differential Privacy.

Lab 6 → Neural Networks → Perceptron
Backpropagation
Convolutional Neural Networks

Lab 7 → Transformers → Transformer
Self-Attention
Vector Database. → ^{Supplement} QKV
word embeddings
↳ 7.1 → Logit Lens (auditing LLMs)

Lab 8 → Gentle Intro to LLMs → Huggingface 10 NLP tasks
↳ for loops

Lab 9 → LLMs with APIs → API
→ Prompt engineering

Lab 10 → LLMs with RAGs → RAGs → ^{Plus}

Lab Bonus (self-learning) → LLM APPs / Interfaces → context injection Graph RAG
→ Gradio, end-to-end solutions